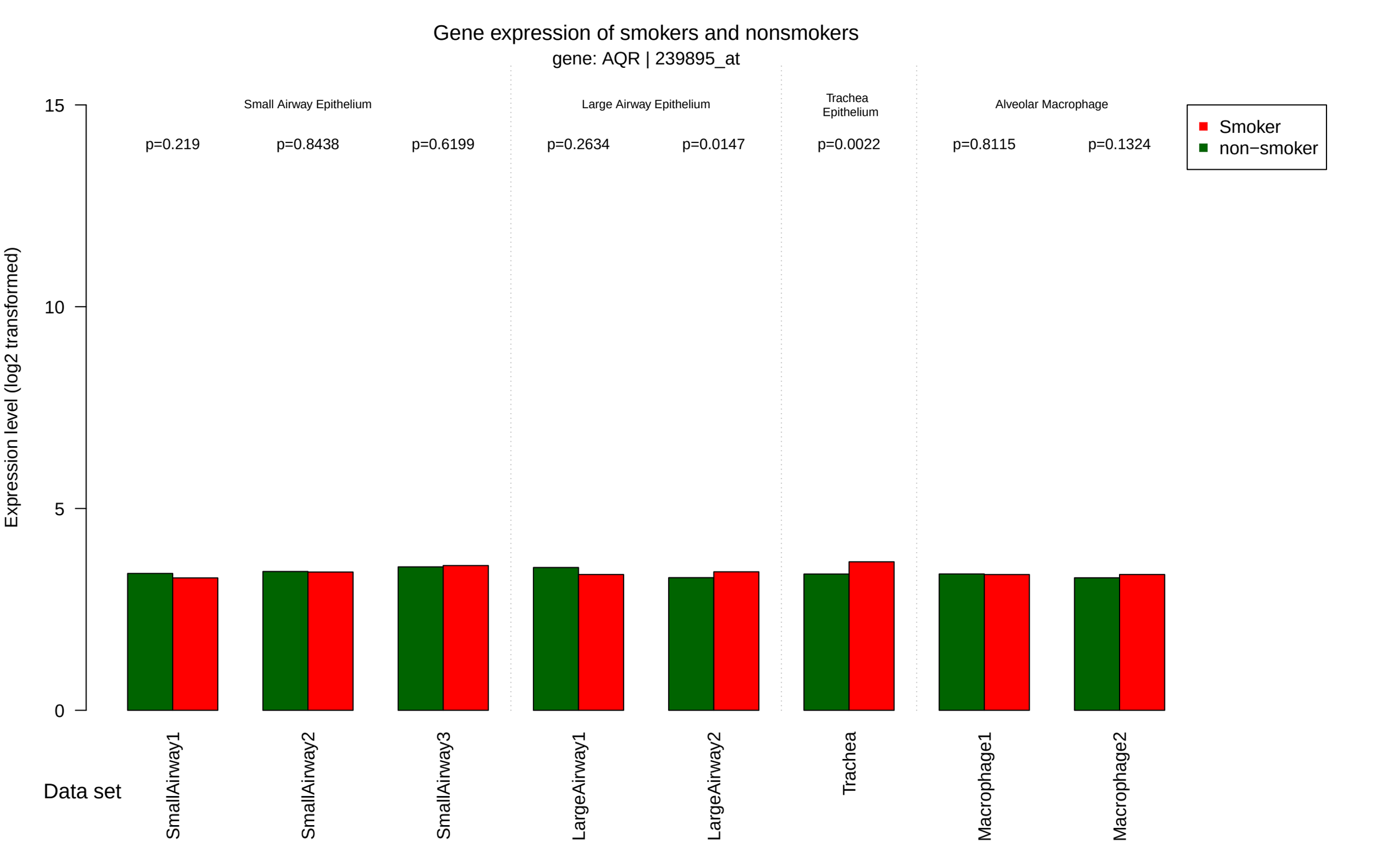
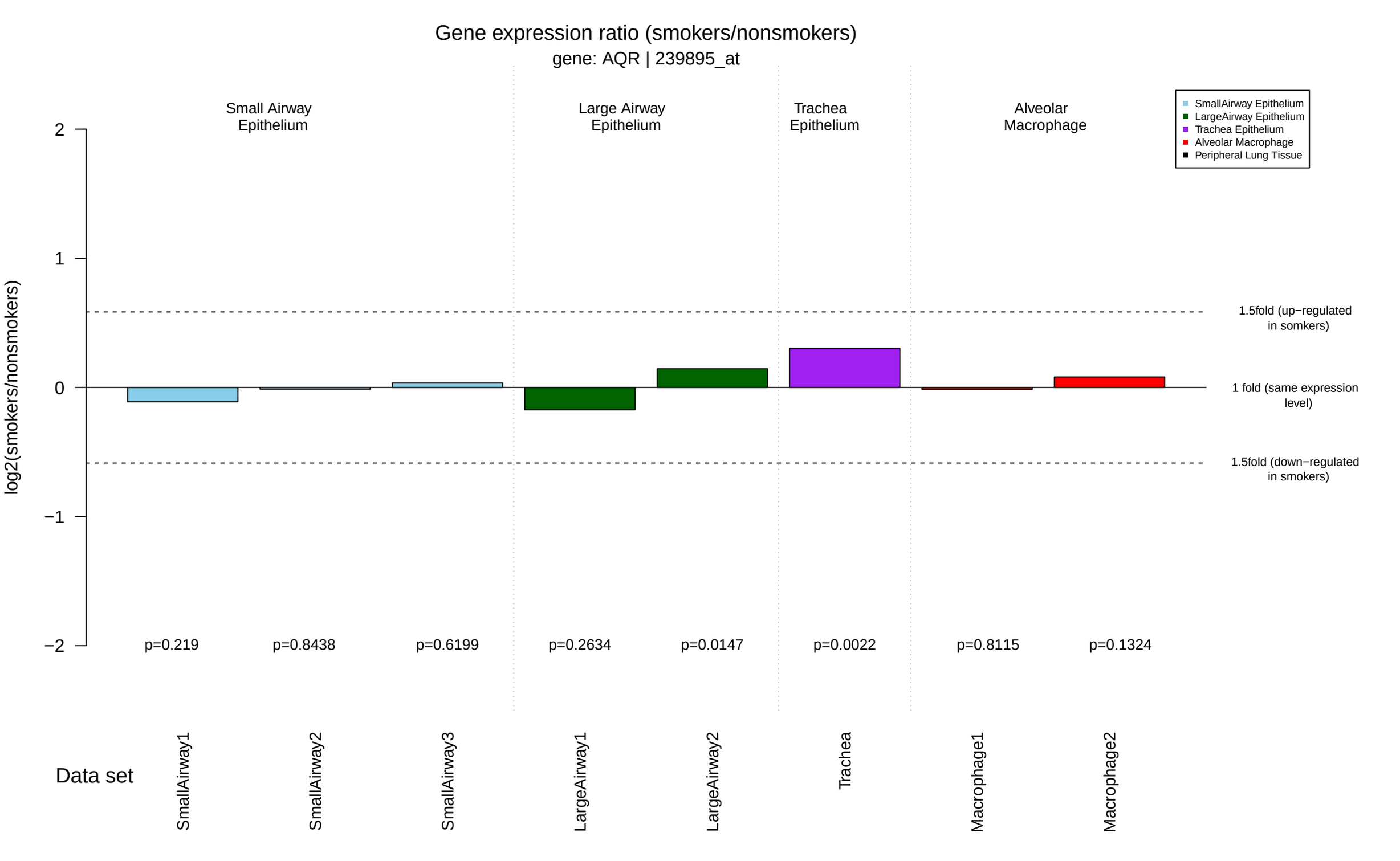
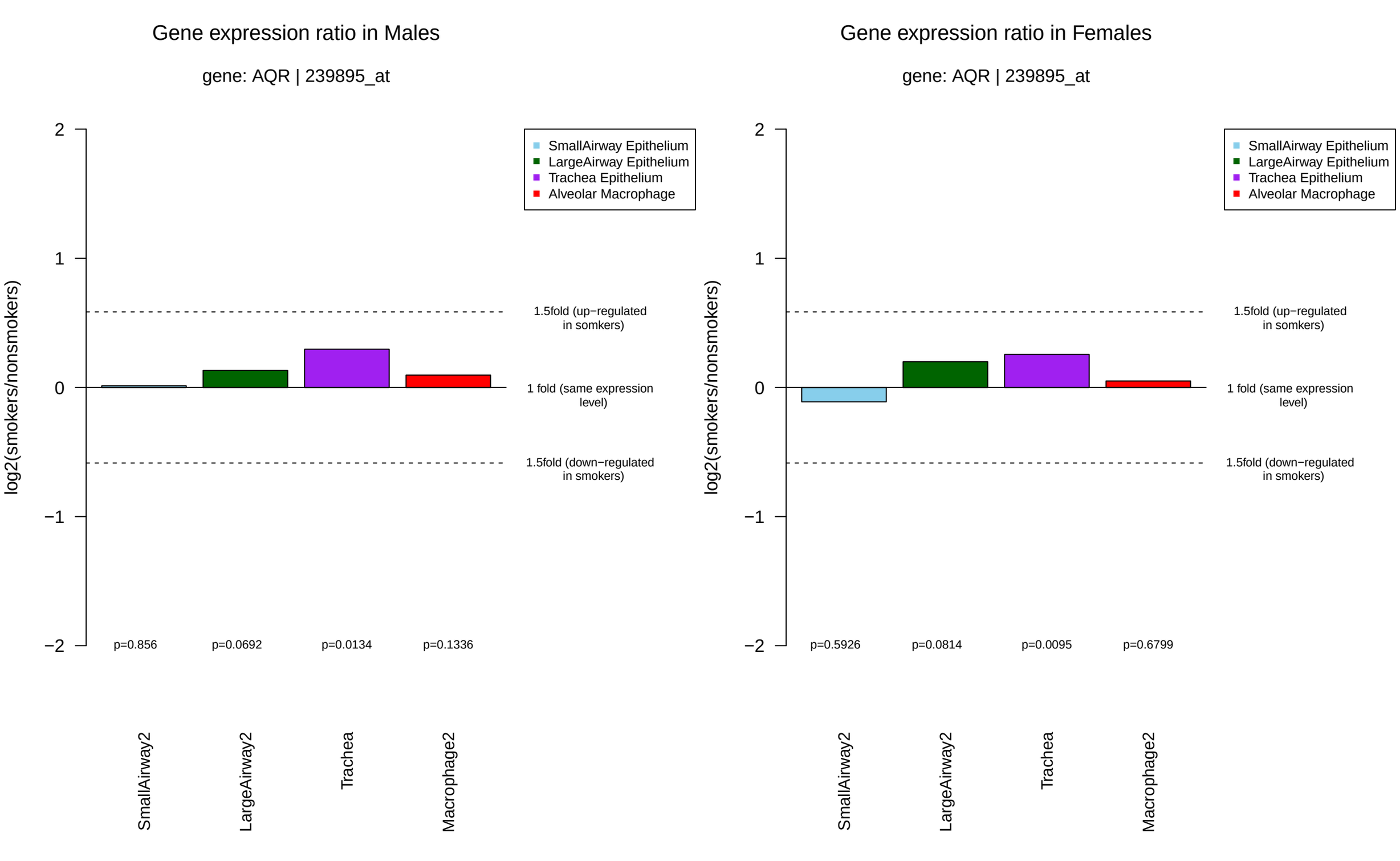
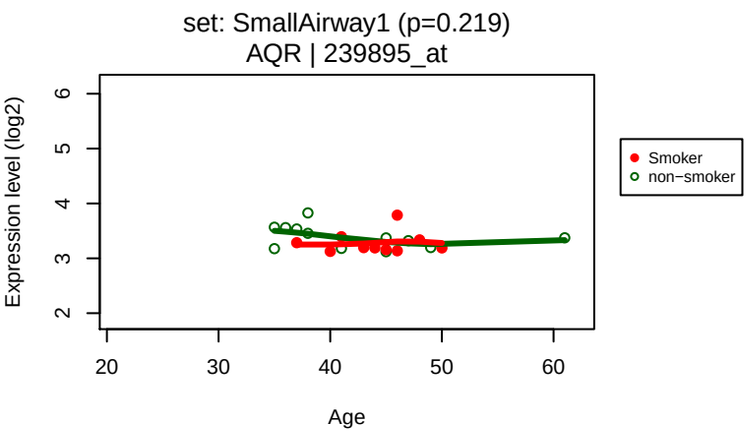
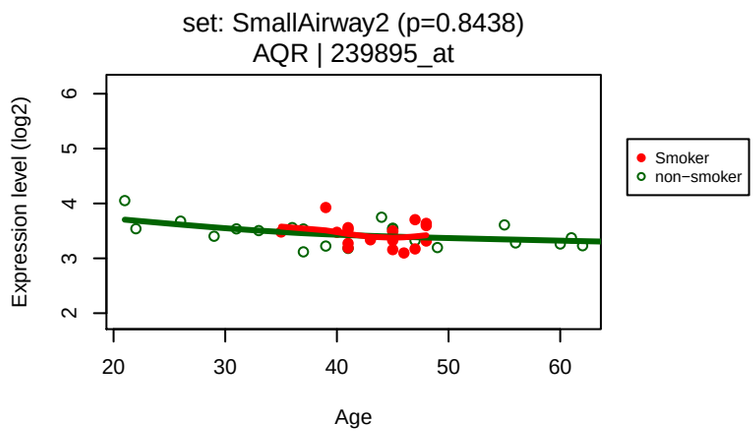
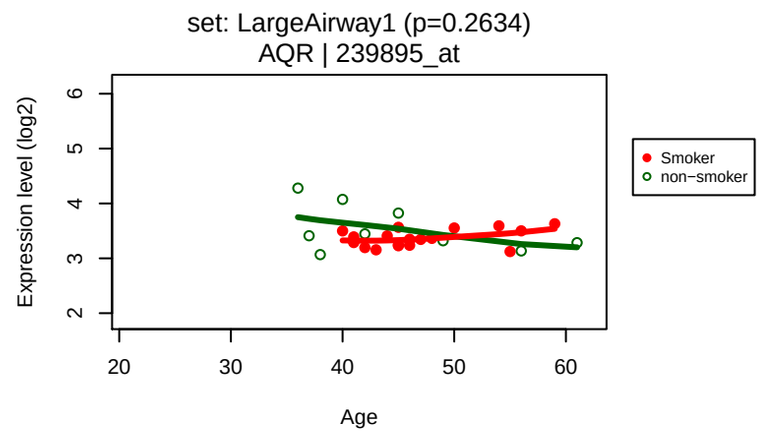

Gene expression of smokers and nonsmokers
gene: AQR | 239895_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.219
p=0.8438
p=0.6199
p=0.2634
p=0.0147
p=0.0022
p=0.8115
p=0.1324
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

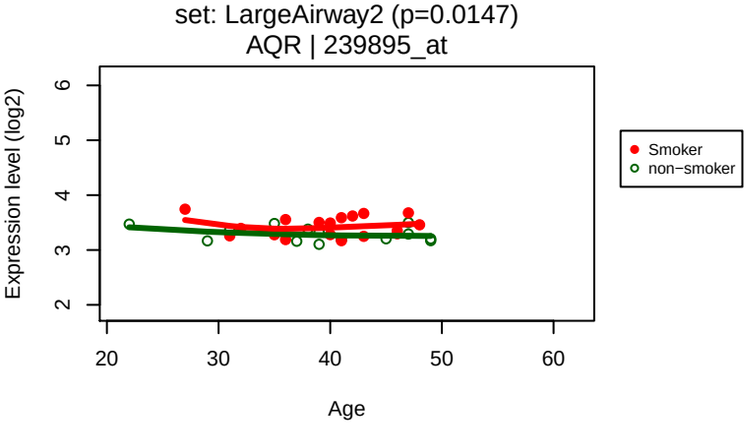

Gene expression ratio (smokers/nonsmokers)
gene: AQR | 239895_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.219
p=0.8438
p=0.6199
p=0.2634
p=0.0147
p=0.0022
p=0.8115
p=0.1324
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

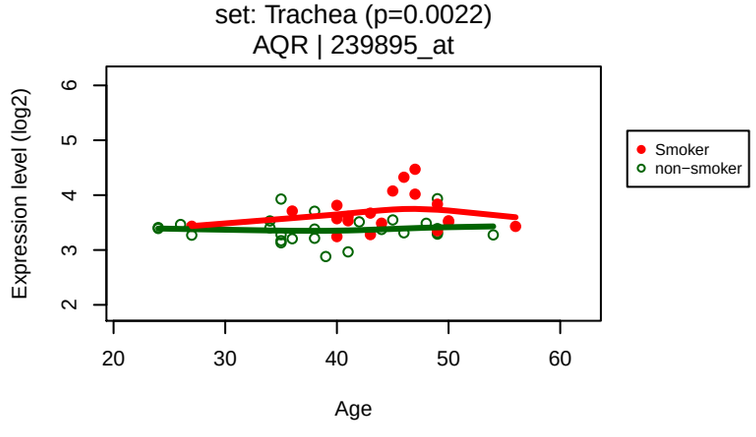

Gene expression ratio in Males
gene: AQR | 239895_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.856
p=0.0692
p=0.0134
p=0.1336
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQR | 239895_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5926
p=0.0814
p=0.0095
p=0.6799
SmallAirway2
LargeAirway2
Trachea
Macrophage2

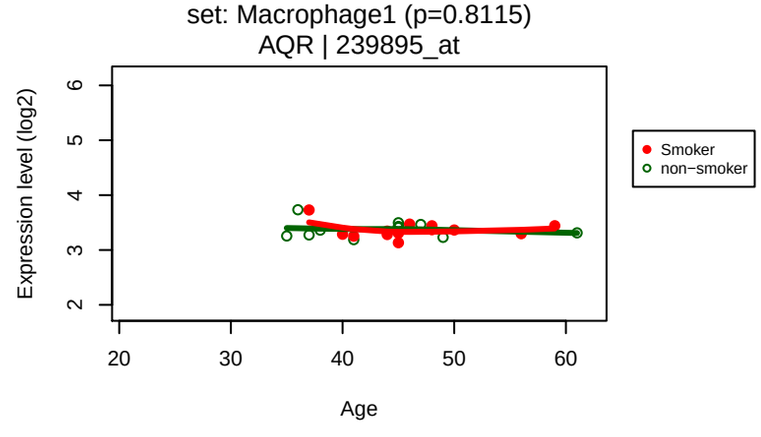

set: SmallAirway1 (p=0.219)
AQR | 239895_at
Expression level (log2)
Age
Smoker
non−smoker

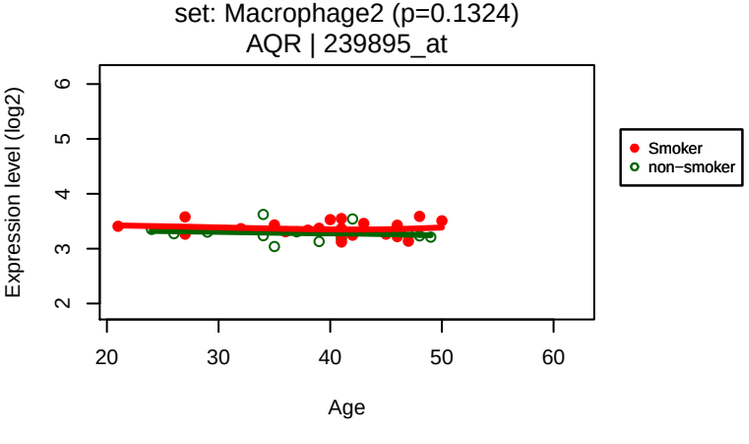

set: SmallAirway2 (p=0.8438)
AQR | 239895_at
Expression level (log2)
Age
Smoker
non−smoker

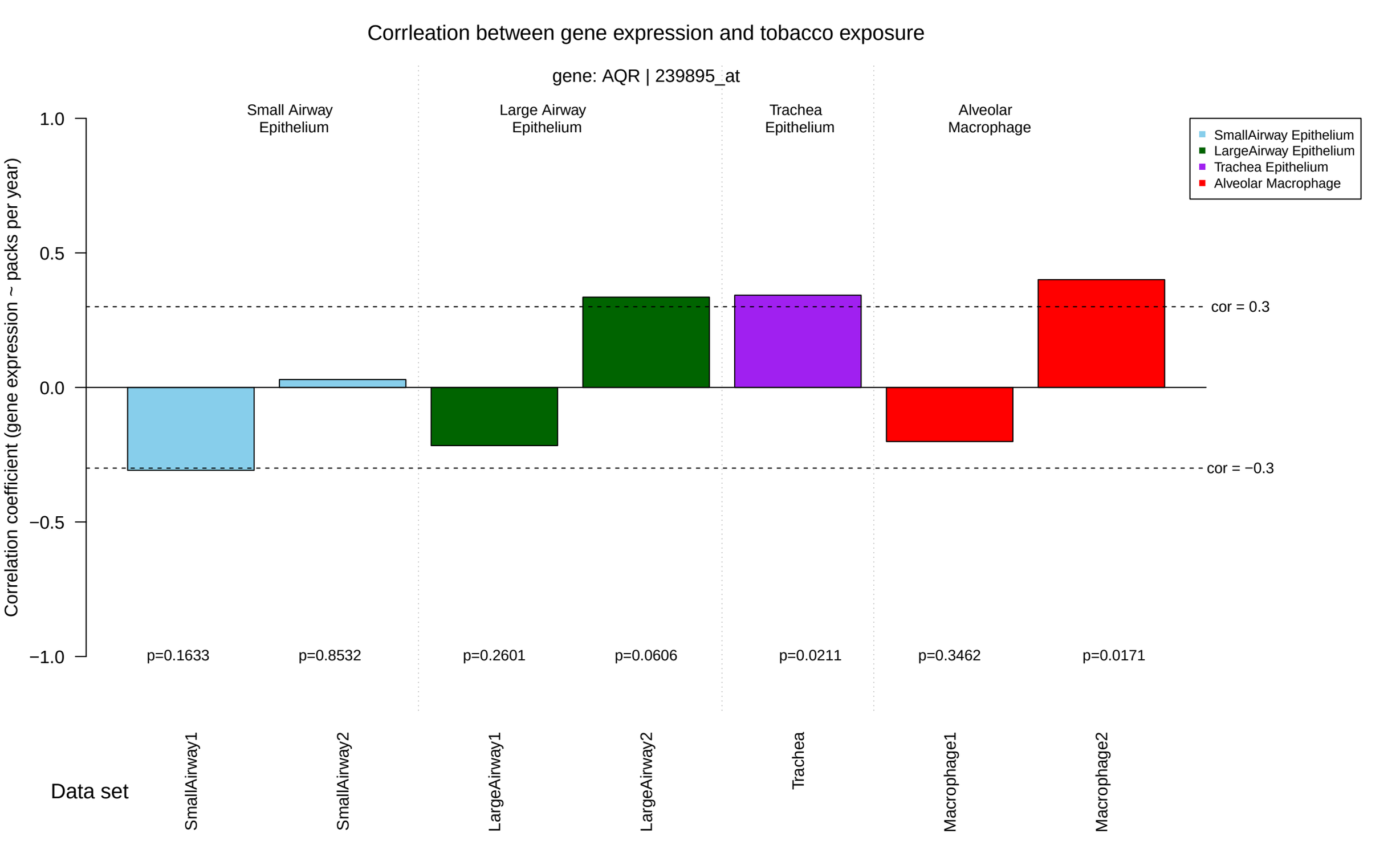

set: LargeAirway1 (p=0.2634)
AQR | 239895_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0147)
AQR | 239895_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0022)
AQR | 239895_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8115)
AQR | 239895_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1324)
AQR | 239895_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQR | 239895_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1633
p=0.8532
p=0.2601
p=0.0606
p=0.0211
p=0.3462
p=0.0171
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2